

- ① Find optimal straight line, crossroads and corner PID gains & velocity target (manual testing)
 - ② Collect downward camera data automatically using PID controller and categorise
- 1
- ③ Create CNN to learn how to categorise image and update PID gains & speed
- 2

④ Compare models

straight corner crossroad

$v = 20 \text{ m/s}$

$k_p = 2$

$k_I = 3$

$k_D = 2$

$v = 5 \text{ m/s}$

$k_p = 3$

$k_I = 5$

$k_D = 4$

$v = 3 \text{ m/s}$

$k_p = 5$

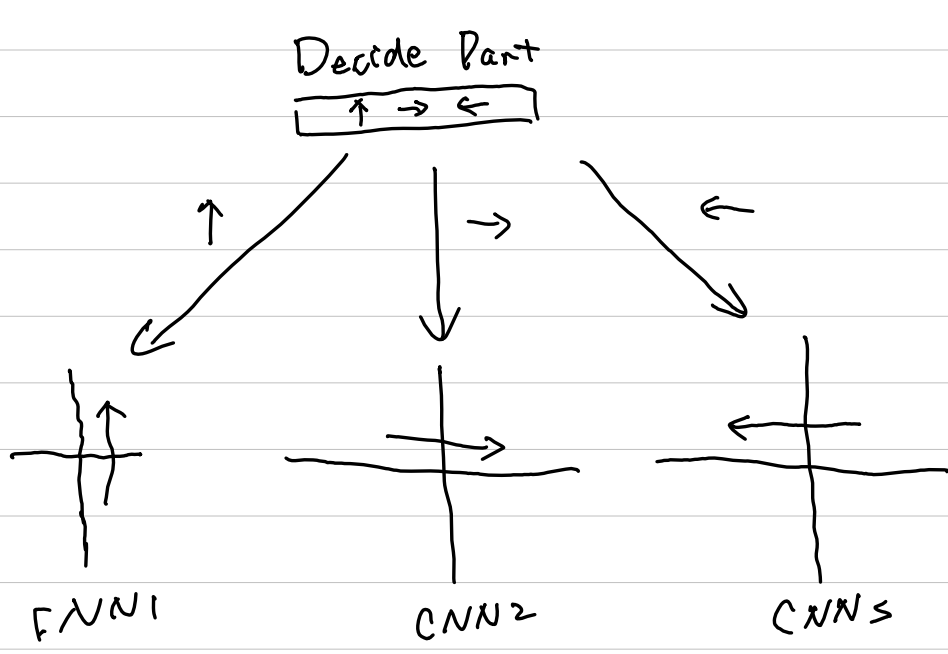
$k_I = 3$

$k_D = 7$

yaw rate = 4 rad/s

target

CNN Classifications



1

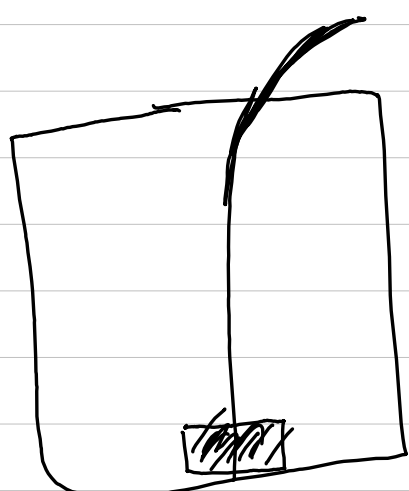
2

Nev
Yaohong

Karthik
Yunkao Huang

Xiaowei

Tao Bai



straight $\rightarrow$ corner soon?

Straight

Corner

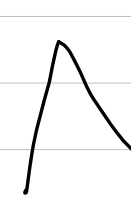
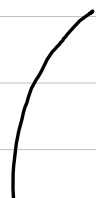
Straight soon

corner



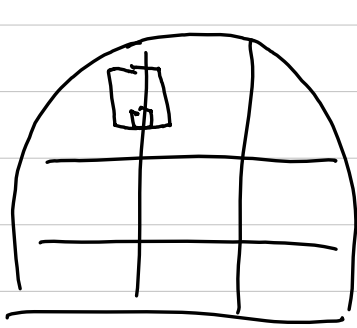
Easy Corner

Hard Corner



CNN $\rightarrow$ 3?

6?



undistorted_image

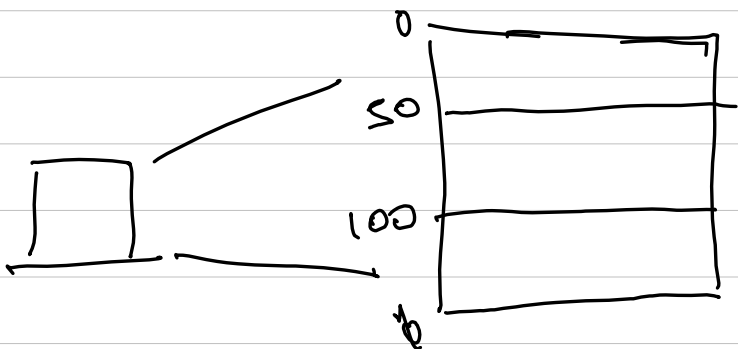
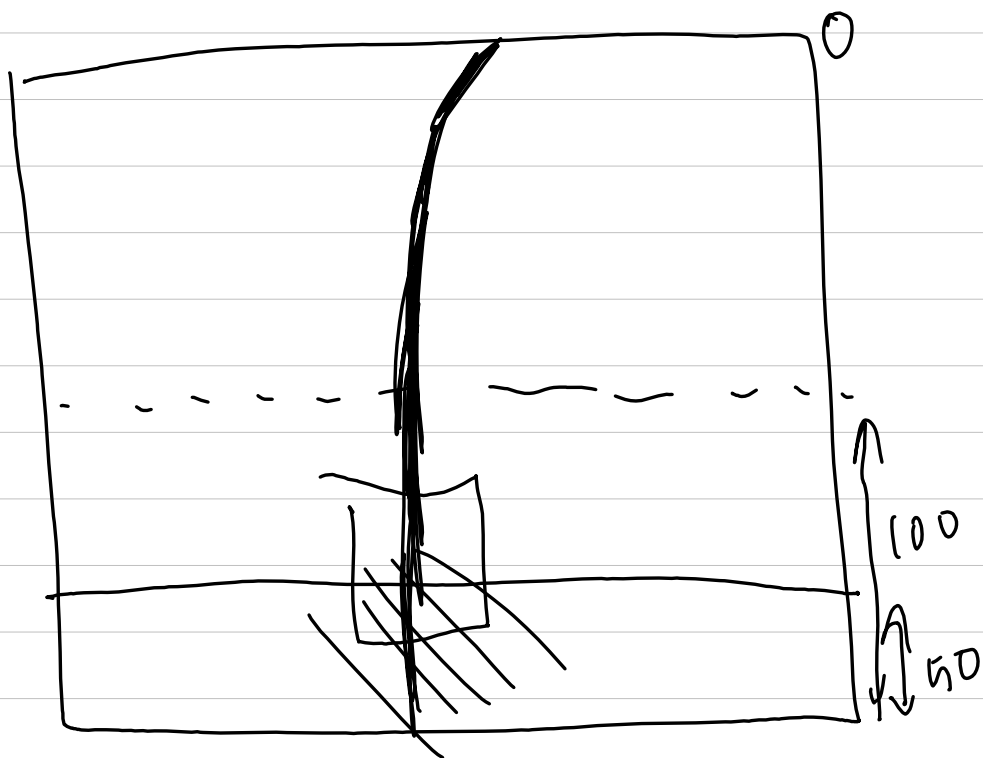
undistorted_image(1)

undistorted_image(2)

undistorted_image(3)

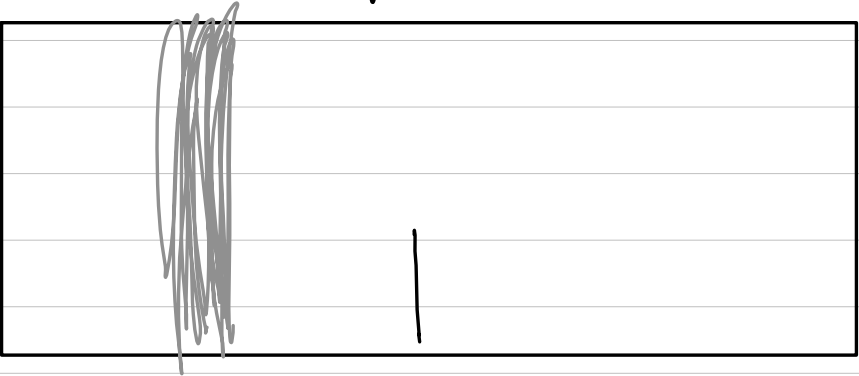
gray scale

gray scale vs full image





Centre ✓ → Fast
Straight ✓ For Spd = 1



Side → slow
Straight For Spd = 0.1



Straight
Curve For Spd = 0.4

Velocity



